

APM1110_FA6

https://github.com/DyasMine-MD/APM1110-FA6/blob/main/AMP1110_FA6_DICHE_JASMINE

Jasmine Diche

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I. Geometric Distribution

The geometric distribution models the number of trials required to achieve the first success in a sequence of independent Bernoulli trials, each having a constant probability of success p .

Note on R's Implementation: R's `rgeom(n, p)` counts failures before the first success. To model the **total number of trials** required, we add 1: `rgeom(n, p) + 1`.

1. Set the probability of success

```
p <- 0.2
```

2. Generate 1000 random variables (trials to first success)

```
set.seed(42) # For reproducibility
x <- rgeom(1000, prob = p) + 1
```

3. Calculate basic statistics

```
mean_x <- mean(x)
var_x <- var(x)
sd_x <- sd(x)
```

4. Print formatted results

```
cat("Number of trials required to achieve first success:\n")
```

```
## Number of trials required to achieve first success:
```

```
cat("Mean (in 2 decimal places):", sprintf("%.2f", mean_x), "\n")
```

```
## Mean (in 2 decimal places): 5.16
```

```
cat("Variance (in 2 decimal places):", sprintf("%.2f", var_x), "\n")
```

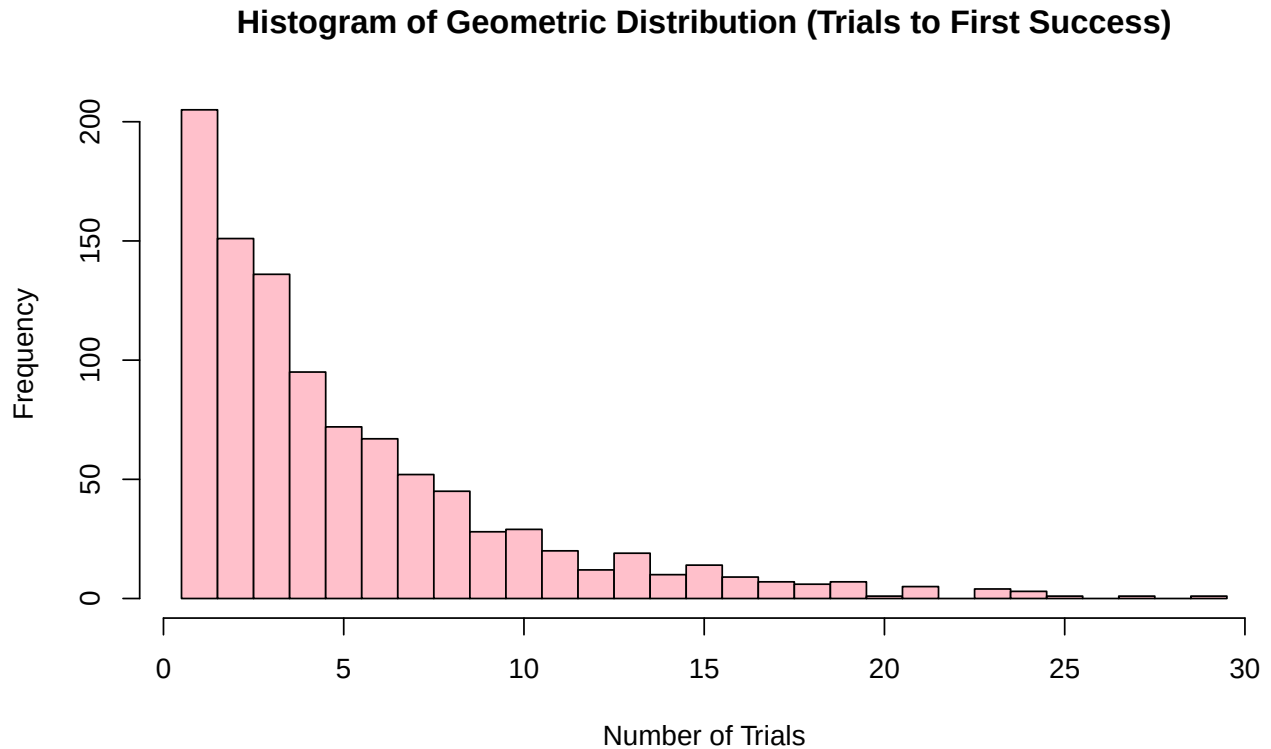
```
## Variance (in 2 decimal places): 21.69
```

```
cat("Standard deviation (in 2 decimal places):", sprintf("%.2f", sd_x), "\n")
```

```
## Standard deviation (in 2 decimal places): 4.66
```

5. Histogram of Results

```
hist(x,
     breaks = seq(0.5, max(x) + 0.5, by = 1),
     col = "pink",
     border = "black",
     main = "Histogram of Geometric Distribution (Trials to First Success)",
     xlab = "Number of Trials",
     ylab = "Frequency")
```



II. Hypergeometric Distribution

A plant manufactures IC chips of which 10% are expected to be defective. A sample of size $n = 10$ is drawn from a box before transportation. We want to estimate the probability that the sample contains **more than 10% defectives**.

For a sample size of $n = 10$, 10% corresponds to $10 \times 0.10 = 1$ defective chip. Therefore, containing *more than 10% defectives* means finding **more than 1 defective chip** ($X > 1$, or $X \geq 2$).

1. Sample of 10 selected from a box of 40 ($N = 40$)

- Population size: $N = 40$
- Defective chips in population: $m = 40 \times 0.10 = 4$
- Non-defective chips in population: $n_{non} = 40 - 4 = 36$
- Sample size: $k = 10$

```
# Parameters for Box of 40
N1 <- 40
m1 <- 4      # 10% defective
n1 <- N1 - m1 # 36 non-defective
```

```

k_sample <- 10

# Probability of > 10% defectives
p1 <- phyper(1, m = m1, n = n1, k = k_sample, lower.tail = FALSE)

# Mathematical steps
px0_1 <- dhyper(0, m = m1, n = n1, k = k_sample)
px1_1 <- dhyper(1, m = m1, n = n1, k = k_sample)

cat("P(Defectives > 10% | N = 40) =", round(p1, 4), "\n")

## P(Defectives > 10% | N = 40) = 0.2559

Calculations: *  $P(X = 0) \approx 0.2999$  *  $P(X = 1) \approx 0.4442$  *  $P(X > 1) = 1 - (0.2999 + 0.4442) = \mathbf{0.2559}$ 

```

2. Sample of 10 selected from a box of 5000 ($N = 5000$)

- Population size: $N = 5000$
- Defective chips in population: $m = 5000 \times 0.10 = 500$
- Non-defective chips in population: $n_{non} = 5000 - 500 = 4500$
- Sample size: $k = 10$

```

# Parameters for Box of 5000
N2 <- 5000
m2 <- 500      # 10% defective
n2 <- N2 - m2  # 4500 non-defective

# Probability of > 10% defectives
p2 <- phyper(1, m = m2, n = n2, k = k_sample, lower.tail = FALSE)

# Mathematical steps
px0_2 <- dhyper(0, m = m2, n = n2, k = k_sample)
px1_2 <- dhyper(1, m = m2, n = n2, k = k_sample)

cat("P(Defectives > 10% | N = 5000) =", round(p2, 4), "\n")

## P(Defectives > 10% | N = 5000) = 0.2639

Calculations: *  $P(X = 0) \approx 0.3483$  *  $P(X = 1) \approx 0.3878$  *  $P(X > 1) = 1 - (0.3483 + 0.3878) = \mathbf{0.2639}$ 

```